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Bill Scanning Based Reward & Intelligence System

Study Project Report

Phase "1"

Presented By **Team ARAJ** – Group (120)
Advised By **Prof. Uma Sankara Rao**

Project Details:

Course Title	B.Sc. Computer Science – BITS Pilani Digital	
Project Title	Bill Scanning Based Reward & Intelligence System	
Phase	1	
Group No.	120	
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Submission Date	04 | February | 2026	

Team ARAJ

Who We Are

Team ARAJ, Group 120, is a four-member student team from the B.Sc. Computer Science program (BITS Pilani Digital). This study project is completed as part of our final-year academic requirements, emphasizing real-world problem identification and the creation of feasible, technology-driven solutions.

Our chosen project area is Artificial Intelligence and Machine Learning (AI/ML), focusing on data extraction, pattern recognition, and converting unstructured data into meaningful, privacy-compliant intelligence.

Bill Scanning–Based Reward & Intelligence System

Team ARAJ — **A**shfaaq, **R**anjeet, **A**rpan, **J**yoti

Phase 1: Problem Identification & Planning

Problem Statement & Motivation

Problem Statement

Consumers produce vast amounts of transactional data through their everyday bills and receipts, spanning categories like groceries, utilities, travel, electronics, and services. Unfortunately, this data is often fragmented, unstructured, and left unused after the transaction concludes. Meanwhile, brands do not have access to valuable cross-platform, purchase-level intelligence that reflects actual consumer behavior outside their own ecosystems.

Existing reward and cashback systems are typically brand-specific, siloed, and limited in the insights they provide. These systems fail to capture a comprehensive view of consumer spending across different merchants, categories, and time periods.

Summary

Consumers generate valuable bill data that remains fragmented and unused, while brands lack cross-platform purchase insights. Existing reward systems are siloed, brand-specific, and fail to capture holistic consumer spending behavior.

Motivation

The primary motivation for this project is to design a mutually beneficial system where:

- **Users** receive tangible rewards for scanning bills they already have.
- **Brands and partners** obtain access to privacy-compliant, aggregated purchase intelligence, enabling them to better understand consumer spending patterns, preferences, and trends.

By transforming raw bill images into anonymized behavioral insights, the system unlocks value from data that would otherwise be wasted.

Literature Review / Background Study

Receipt Scanning & OCR

Advances in **Optical Character Recognition (OCR)** and computer vision have made it possible to automatically extract structured data from invoices and receipts. Tools such as expense trackers and accounting applications utilize OCR for digitizing bills, but their focus remains on individual record-keeping rather than producing large-scale intelligence. [1]

Consumer Analytics & Behavioral Data

Modern consumer analytics primarily relies on first-party and third-party data, which is often restricted to behavior observed within specific platforms, such as e-commerce browsing. Access to cross-merchant, purchase-level data is rare, expensive, and typically gathered through surveys or panel data, limiting both accuracy and scale. [2]

Reward-Driven Data Sharing

Research indicates that users are more willing to share personal data when they perceive a clear value exchange, such as cashback, discounts, or loyalty rewards. Incentivized data contribution models have demonstrated success in areas like fintech and survey platforms. [3]

Gap Identified

Current systems are unable to:

- Aggregate **cross-category, real purchase data**
- Provide **continuous, scalable insights**
- Balance **user privacy with commercial intelligence**

This project aims to address these gaps.

Existing System Analysis

Existing Approaches

Brand-Specific Loyalty Programs

1. Limited to a single merchant or brand
2. No cross-category visibility
3. Data remains locked within the brand ecosystem

Expense & Bill Tracking Apps

1. Focused on personal finance management
2. No reward incentives
3. No intelligence layer for brands

Coupon & Cashback Platforms

1. Primarily transaction-triggered or affiliate-based
2. Minimal understanding of users' holistic spending behavior

Limitations of Existing Systems

- Fragmented data silos
- No standardized purchase-level analytics
- Low user engagement after initial use

Commercial & Industry Systems

- **Expensify SmartScan** [4]

  A popular receipt scanning workflow where users capture receipts and the system extracts expense details automatically. This demonstrates real-world viability of OCR-based expense extraction pipelines.

- **Amazon Textract – Analyze Expense** [5]

  A dedicated API for invoices/receipts that returns structured JSON including summary fields and line item groups. Shows a production-grade approach to receipt understanding beyond plain OCR.

- **Microsoft Azure Document Intelligence** [6]

  A prebuilt receipt model combining OCR and deep learning to extract key receipt fields. Demonstrates strong baseline capability for field extraction in varied receipt formats.

- **Fetch Rewards** [7]

  a similar company, who got an investment of $590M USD, in 14 rounds and is valued at $2.9B

- **Receipt Scanning Reward Based Market info** [8]

- **With the fall of Cookie based browsers and the decline of cookie based advertising, the need for First Party Data for DSPs is higher than ever.** [9]

Proposed Solution (High-Level)

Solution Overview

The proposed system allows users to **scan bills from any merchant** and receive **instant rewards** such as cashback, coupons, or offers. Simultaneously, the platform converts the extracted data into **privacy-compliant, anonymized intelligence** .

In addition to physical bills, users also generate significant amounts of transactional data through SMS notifications and OTP messages. Furthermore, users frequently receive digital bills and receipts via email. With user consent, the system can be granted automated access—restricted solely to emails with bill-related subjects or PDF receipts—for seamless extraction of digital bill data. Alternatively, users may choose to manually upload digital bills, ensuring flexibility and control over their data sharing preferences.

Summary

The system enables users to scan physical or digital bills to earn instant rewards, while converting data into privacy-compliant intelligence. With user consent, bill data can be extracted from SMS, emails, or manual uploads.

ML Pipeline

OCR → Cleaning → Categorization → Insights

OCR / Receipt Extraction

Extract text blocks, words, bounding boxes, and confidence scores.

If using managed services:

- AWS Textract AnalyzeExpense is designed for invoices/receipts and returns structured expense objects.
- Azure Document Intelligence Receipt model extracts receipt fields (merchant name, date, total, tax) as structured JSON.
- Google Vision OCR provides text detection/extraction on images.

Cleaning & Normalization

- **Fix OCR artifacts:**
  currency symbols, decimal separators, merged tokens, date formats.
- **Normalize merchants using:**
  Regex cleanup + fuzzy match + embedding similarity
- **Validate totals:**
  reconcile subtotal + tax ≈ total where possible (tolerance threshold).

Categorization & Parsing

- **Extract features:**
  merchant string, item keywords, quantities/prices, inferred category terms
- **Model options:**
  Baseline: rules + dictionaries (fast, interpretable**)**
  ML: supervised classifier for category prediction
  KIE: for line-items and key-value fields (SROIE/CORD-aligned).

Insights Generation

- **Per-user analytics (private view):**
  monthly spend by category, merchant preference, top items
- **Aggregated intelligence (anonymized):**
  category-level trends, brand affinity, cohort segmentation
- **Predictive layer (Optional later):**
  churn/retention likelihood, offer response propensity.

Objectives, Scope & Constraints

Objectives

- Enable users to monetize their bill data through rewards
- Generate cross-category, purchase-level intelligence
- Maintain strict privacy and anonymization standards
- Create a scalable, integration-friendly system

Scope

Included

- Bill scanning and data extraction
- Basic categorization (e.g., groceries, travel, utilities)
- Reward allocation logic
- High-level analytics and insights
- SDK-based integration model

Excluded (Phase 1)

- Real-time merchant integrations
- Advanced predictive analytics
- Full-scale monetization with enterprise brands

Constraints

- Data accuracy depends on bill image quality
- Initial insights limited by sample size
- Regulatory and privacy compliance requirements
- Dependence on partner adoption for monetization

Conclusion

This proposal presented a comprehensive study of a **Bill Scanning–Based Reward and Intelligence System**, addressing a real-world gap in how consumer purchase data is captured, utilized, and monetized. The report initially outlined the problem context, motivation, identified gaps, solution overview, system components, deployment model, objectives, scope, and constraints, clearly demonstrating the feasibility and relevance of the proposed system within a B.Sc. Computer Science study project framework.

Building upon this foundation, the extended technical design introduced a clear AI/ML-driven architecture, detailing the model types, data flow, and end-to-end machine learning pipeline from receipt capture and OCR, through data cleaning and categorization, to anonymized intelligence generation. The inclusion of dataset selection strategies, evaluation metrics, and a deeper literature review referencing both industry systems and academic benchmarks strengthens the project's academic rigor and aligns it with standard research and engineering practices.

Overall, the combined proposal establishes a scalable, privacy-aware, and technically grounded system that not only rewards users for data contribution but also transforms fragmented receipt data into meaningful, aggregated insights. The project is well-positioned for Phase-I implementation and validation, with clear pathways for future enhancement in predictive analytics, personalization, and real-time integrations, making it suitable for both academic evaluation and potential real-world extension.

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[4] **Expensify SmartScan:**
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[5] **Amazon Textract – AnalyzeExpense:**
https://docs.aws.amazon.com/textract/latest/dg/analyzing-document expense.html

[6] **Microsoft Azure Document Intelligence – Receipt Model:**
https://learn.microsoft.com/en-us/azure/ai-services/document intelligence/prebuilt/receipt?view=doc-intel-4.0.0

[7] **Fetch Rewards**

- https://www.axios.com/sponsored/how-rewards-turn-advertising-into-a-win-win-for-brands-and-consumers

[8] **Receipt Scanning Reward Based Market info**

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[9] **With the fall of Cookie based browsers and the decline of cookie based advertising, the need for First Party Data for DSPs is higher than ever.**

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The End
Thank You..

Submission Date	Group Advisor	TEAM
04-Feb-2026	Prof. Uma Sankara Ra	ARAJ